A Project Report on

Academic Project Selection
Department of CSE

submitted in partial fulfillment for the award of

Bachelor of Technology
in
Computer Science & Engineering
By

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CERTIFICATE

This is to certify that the project report entitled **Academic Project Selection** that is being submitted by L.Sivamani(Y22ACS489), K.Y.Sai Sandeep (L23ACS604), M.Gayathri(Y22ACS502), Y.Vamsi Krishna (Y22ACS595) in partial fulfillment for the award of the Degree of Bachelor of Technology in Computer Science & Engineering to the Acharya Nagarjuna University is a record of bonafide work carried out by them under our guidance and supervision.

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DECLARATION

We declare that this project work is composed by ourselves, that the work contained herein is our own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

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Table Of Contents

List Of Figures viii

List Of Tables ix

List Of Abbreviations x

Abstract xi

1 Introduction 1

1.1 Motivation 1

1.2 Problem Statement 2

1.3 Objective of Project 3

2 Literature Survey 4

2.1 Existing System 6

2.2 Disadvantages of Existing System 6

2.3 Proposed System 7

2.4 Advantages of the Proposed System 8

3 System Analysis 10

3.1 Software Requirement Specification 11

3.1.1 User Requirements 11

3.1.2 Software Requirements 12

3.1.3 Hardware Requirements 13

3.1.4 Libraries and Tools Used 14

3.2 Content Diagram of Project 15

3.2.1 Diagram Explanation 16

4 Design of the Project 18

4.1 Introduction 18

4.2 DFD / ER Diagrams 18

4.2.1 DFD Level 0 19

4.2.2 DFD Level 1 20

4.2.3 ER Diagram 22

4.3 Data Dictionary 25

5 Implementation and Results 29

5.1 Method of Implementation 30

5.1.1 Implementation of Algorithms 32

5.1.2 Methodology of the Project 39

5.1.3 Proposed Methodology 39

5.1.4 Approach of the Project 41

5.1.5 Output Screens 43

5.1.6 Result Analysis 51

6 Testing and Validation 53

6.1 Features to be Tested 54

6.2 Features not to be Tested 55

6.3 Testing Tools and Environment 57

6.4 Design of test cases and scenarios 58

7 Conclusion and Future Enhancements 60

8 Bibliography 62

List Of Figures

Figure 3.2.1 Top-Level Structure 15

Figure 3.2.2 Flow of Content 16

Figure 4.2.1 DFD Level 0 19

Figure 4.2.2 DFD Level 1 20

Figure 4.2.3 ER Diagram 22

Figure 5.1.1 Work Flow 37

List Of Tables

Table 3.1.1 Software Requirements 12

Table 3.1.2 Hardware Requirements 13

Table 4.1.1 Users Table 25

Table 4.2.2 Projects Table 26

Table 4.2.3 Similarity Results Table 27

Table 4.2.4 Reviews Table 28

List Of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
ASGI	Asynchronous Server Gateway Interface
CORS	Cross-Origin Resource Sharing
CSE	Computer Science and Engineering
DFD	Data Flow Diagram
ER	Entity-Relationship
FastAPI	Fast Application Programming Interface (Python Web Framework)
HTTP	Hypertext Transfer Protocol
JWT	JSON Web Token
LLM	Large Language Model
NLP	Natural Language Processing
ORM	Object-Relational Mapping
RBAC	Role-Based Access Control
REST	Representational State Transfer
SRS	Software Requirements Specification
TF-IDF	Term Frequency – Inverse Document Frequency
UI	User Interface
UX	User Experience
VITE	Vite Build Tool (Next-Generation Frontend Tooling)

Abstract

The **Academic Project Selection System** is a web-based application developed to automate and improve the process of project submission, evaluation, and approval in academic environments. In traditional systems, project selection is handled manually, which often leads to issues such as duplication of project topics, increased workload on faculty members, lack of proper monitoring, and inefficiency in handling large volumes of student submissions.

To overcome these limitations, the proposed system introduces an intelligent and automated approach by integrating modern technologies such as AI and NLP. The system provides an online platform where students can submit their project proposals along with relevant details. Once submitted, the system processes the content and performs similarity analysis using techniques like TF-IDF and cosine similarity to identify duplicate or closely related project ideas.

In addition to similarity detection, the system incorporates AI-based evaluation to assess the quality, clarity, and completeness of the submitted project proposals. This helps in providing meaningful feedback to students and assists faculty members in making informed decisions. The system also ensures a structured workflow by allowing guides and administrators to review, approve, or reject project submissions efficiently.

The application is designed with role-based access control, ensuring that different users such as students, guides, and administrators have appropriate permissions. It also maintains a centralized database for storing project information, making it easy to retrieve, update, and manage records securely.

Furthermore, the system enhances transparency and accountability by maintaining logs of submissions and decisions. It reduces manual effort, minimizes errors, and speeds up the entire project selection process. The scalable design of the system allows it to handle a large number of users and project submissions effectively.

In conclusion, the Academic Project Selection System provides a reliable, efficient, and intelligent solution for managing academic projects. It ensures originality of ideas, improves evaluation accuracy, and significantly reduces the workload on facul-

ty, thereby enhancing the overall efficiency of the academic project management process.

Keywords: Academic Project Selection, Artificial Intelligence (AI), Natural Language Processing (NLP), TF-IDF, Cosine Similarity, Web Application, Project Management System, Similarity Detection, AI-Based Evaluation, Automation.

1 Introduction

The **Academic Project Selection System** is a web-based application developed to simplify and automate the process of managing final year student projects in academic institutions. In colleges, students often submit project ideas that are similar or duplicate to previously submitted projects without knowing. Faculty members and guides spend a significant amount of time manually reviewing and comparing projects, which is both time-consuming and inefficient.

Project Vault addresses this problem by automatically detecting duplicate or similar projects using three NLP engines: spaCy Semantic similarity, NLTK-based TF-IDF, and Jaccard Keyword similarity. When a student submits a project, the system compares it with all existing projects stored in the database and calculates a similarity score. If the score is below the 55% threshold, the project is considered unique, and the student is allowed to proceed with registration. If the score exceeds 55%, the student is notified and required to revise the project.

The system also incorporates AI-based validation using Groq AI powered by the LLaMA 3 model to assess the quality and meaningfulness of project content before submission. This ensures that students do not submit irrelevant or meaningless content.

Project Vault is developed using modern technologies, with FastAPI for the backend, React with Vite for the frontend, and PostgreSQL as the database. The application is fully deployed on the cloud, with the frontend hosted on Vercel and the backend along with the database hosted on Render.

The system supports three types of users: Student, Guide, and Admin, each with specific roles, permissions, and access levels to ensure efficient project management.

1.1 Motivation

The primary motivation behind developing the Academic Project Selection System is to overcome the limitations of traditional project management methods. In many institutions, project selection is handled manually, which leads to inefficiencies

such as duplication of ideas, lack of proper evaluation, and increased workload for faculty members.

With the rapid advancement of technology, there is a growing need to adopt intelligent systems that can automate and enhance academic processes. By integrating NLP and AI, the system aims to provide a smarter and more efficient solution for managing academic projects.

The system is designed to promote originality by detecting duplicate ideas, improve the quality of submissions through AI-based evaluation, and ensure a transparent and structured workflow. Ultimately, the motivation is to create a system that supports innovation, reduces manual effort, and enhances the overall academic experience for both students and faculty.

1.2 Problem Statement

In academic institutions, final year students submit project proposals without the support of an automated system to verify originality. As a result, faculty members are required to manually review and compare a large number of submissions, which is both time-consuming and prone to human error. Moreover, students lack a mechanism to check whether their project ideas have already been submitted in previous batches, leading to unintentional duplication of projects.

The absence of an intelligent project management system leads to several issues, including repeated project topics, poor quality submissions, and increased workload for faculty members, guides, and administrators. This manual process also reduces efficiency and lacks transparency in the project evaluation workflow.

Therefore, there is a need for an automated and intelligent system that can effectively detect similar or duplicate projects, validate the quality and relevance of submitted content, and streamline the entire project submission, evaluation, and approval process. Such a system would enhance originality, improve efficiency, and reduce the burden on academic staff.

1.3 Objective of Project

The objective of the Academic Project Selection System is to create an efficient digital platform that simplifies the management of final year projects in academic institutions. The system focuses on providing a structured environment where students can submit their project proposals and faculty can review them in an organized manner. By digitizing the process, it eliminates dependency on manual handling and improves overall coordination between students and guides.

Another important objective is to ensure uniqueness in project submissions by identifying similarities between newly submitted and existing projects. The system applies advanced text analysis techniques to compare project content and generate a similarity score, helping in preventing duplication. It also emphasizes maintaining the quality of submissions by validating whether the content is meaningful and relevant before it is accepted into the system.

Additionally, the system aims to support better decision-making for faculty by providing clear insights and automated feedback on project proposals. It creates a centralized repository of project data that can be easily accessed and managed. This not only improves transparency but also helps institutions maintain a record of previous projects for future reference, ultimately enhancing the overall project management process.

2 Literature Survey

The development of intelligent systems for academic project management and validation has evolved significantly with advancements in Natural Language Processing (NLP), machine learning, and web technologies. Early research in text similarity and information retrieval laid the foundation for modern plagiarism detection and content validation systems. Techniques such as Term Frequency–Inverse Document Frequency (TF-IDF) have been widely used to represent textual data numerically and evaluate the importance of words within documents [1].

Salton and Buckley introduced TF-IDF as a fundamental method for information retrieval, enabling efficient comparison of textual documents based on keyword importance. Building upon this, cosine similarity became a standard metric for measuring the similarity between document vectors, providing a normalized score that indicates the degree of resemblance between two texts [2].

Recent advancements in NLP have incorporated semantic analysis to improve accuracy beyond keyword matching. Tools such as spaCy utilize word embeddings and contextual understanding to evaluate the meaning of text rather than just surface-level similarity. This approach significantly enhances the detection of paraphrased or semantically similar content [3]. Additionally, Jaccard similarity has been used for token-based comparison, offering a simple yet effective method for identifying overlapping keywords between documents [4].

In the domain of academic project management systems, researchers have emphasized the importance of automated validation to maintain originality and reduce redundancy. Studies have shown that integrating multiple similarity detection techniques improves overall accuracy and reliability. Hybrid approaches combining TF-IDF, semantic similarity, and token-based methods provide a balanced evaluation of project content [5].

Web-based system implementation has also been widely explored, with frameworks such as Flask and FastAPI enabling efficient backend development for machine learning applications. Pedregosa et al. demonstrated the effectiveness of the scikit-

learn library in implementing machine learning algorithms [6], while modern frontend technologies enhance user interaction through dynamic dashboards and real-time feedback.

Database management systems such as PostgreSQL and SQLite have been proven to efficiently handle structured data related to users, projects, and similarity results. These systems ensure scalability, reliability, and secure storage of academic records [7]. Furthermore, cloud-based deployment has enabled flexible resource management and improved accessibility for users across different platforms.

Overall, the existing body of research highlights the effectiveness of combining NLP techniques, machine learning algorithms, and web technologies to build robust academic project validation systems. These studies collectively justify the design and implementation of the proposed system.

The proposed Academic Project Selection system builds upon these techniques by integrating NLP-based similarity detection and AI-based validation into a unified academic project management platform.

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2.1 Existing System

The existing academic project management systems in most educational institutions are either completely manual or partially digitized. In a typical setup, students submit their project proposals through paper-based formats or basic online forms such as emails or simple portals. Faculty members then manually review these submissions, provide feedback, and approve or reject them.

This traditional approach suffers from several inefficiencies. One of the primary issues is the absence of an automated mechanism to detect duplicate or similar project ideas. As a result, multiple students may unknowingly select similar topics, leading to redundancy and reduced innovation. This not only affects the quality of academic work but also limits students' creativity and exploration of new ideas.

Another major limitation is the time-consuming nature of manual evaluation. Faculty members are required to review each submission individually, which becomes increasingly difficult as the number of students grows. This leads to delays in project approval and increases the workload on faculty.

Additionally, there is a lack of standardized evaluation criteria, which may result in inconsistent judgments. Tracking project progress is also challenging due to the absence of a centralized system. Students often face difficulties in understanding the current status of their submissions, and communication gaps between students and faculty further complicate the process.

Overall, the existing system is inefficient, lacks scalability, and does not support intelligent decision-making, making it unsuitable for modern academic environments.

2.2 Disadvantages of Existing System

- **Manual Process**
  Project verification and approval are done manually, which increases workload for faculty.

- **Time-Consuming**
  Comparing project proposals with previous records takes a lot of time and effort.
- **No Similarity Detection**
  The system does not provide any mechanism to identify duplicate or similar project ideas.
- **High Chances of Duplication**
  Similar or repeated projects may get approved due to lack of proper checking.
- **Error-Prone Process**
  Manual evaluation can lead to human errors and inconsistencies.
- **No Quality Evaluation**
  There is no system to check whether the submitted project content is meaningful or relevant.
- **Lack of Automation**
  Most systems do not use advanced technologies like Artificial Intelligence or Natural Language Processing.
- **Poor Efficiency**
  The overall process is slow and inefficient compared to automated systems.
- **No Centralized Management**
  Project data is not properly organized, making tracking and management difficult.

2.3 Proposed System

The proposed Academic Project Selection System is an intelligent and automated platform designed to overcome the limitations of traditional systems. It is developed as a web-based application that integrates advanced technologies such as NLP and AI to enhance the project selection and evaluation process.

In this system, students can submit their project proposals through an interactive interface. Once submitted, the system automatically processes the data to check for similarities with existing projects using NLP techniques. This ensures that duplicate or closely related project ideas are identified at an early stage.

Furthermore, the system uses AI-based evaluation to assess the quality of project proposals. It analyses various parameters such as clarity, structure, and relevance, and provides constructive feedback to students. This helps in improving the overall quality of submissions.

The system also supports multiple roles, including students, guides, and administrators. Each user has specific responsibilities, ensuring a structured workflow. Guides can review and suggest improvements, while administrators make final decisions regarding approval.

The proposed system improves efficiency, reduces manual workload, enhances transparency, and promotes innovation in academic projects.

2.4 Advantages of the Proposed System

- **Automated Duplicate Detection**
  The system automatically compares every new project against all existing projects in the database using three NLP engines, eliminating the need for manual cross-checks by faculty.
- **Accurate Similarity Detection**
  By combining spaCy Semantic Analysis, NLTK TF-IDF, and Jaccard Keyword algorithms, the system detects similarity even when students paraphrase or reword the same ideas, ensuring robust plagiarism detection.
- **AI-Powered Content Validation**
  Powered by Groq AI using the LLaMA 3 model, the system validates project content for quality and coherence before submission, preventing incomplete or meaningless submissions.
- **Significant Time Savings**
  Both students and guides save valuable time, as the system instantly provides similarity results with a field-wise breakdown—eliminating tedious manual checks.
- **Role-Based Access Control**
  Separate access is provided for Students, Guides, and Admins, ensuring that each user sees only the information relevant to their role.

- **Downloadable Similarity Reports**
  Students can download detailed HTML reports showing field-wise comparison results, which can be submitted along with their project proposals for transparency.
- **Cloud-Based Accessibility**
  Fully deployed on the cloud, the system is accessible from anywhere at any time, without requiring any software installation.
- **Transparent Workflow**
  Every project submission follows a clear, step-by-step process—filling fields, similarity checking, entering team details, and final registration—making the entire workflow transparent and easy to track.

3 System Analysis

System Analysis is a crucial phase in the development of any software system. It in-volves studying the existing system, identifying its limitations, and understanding user requirements to design an improved and efficient solution. This phase helps in defining the overall structure, functionality, and performance requirements of the proposed system.

In the context of the Academic Project Selection System, system analysis focuses on evaluating the current project management practices used in educational institutions. Traditional systems are often manual or semi-automated, leading to inefficiencies such as duplication of project ideas, time-consuming evaluation processes, lack of transparency, and increased workload for faculty members.

The analysis process involves gathering requirements from different including students, guides, and administrators. It identifies key problems such as the absence of similarity detection mechanisms, lack of intelligent evaluation tools, and poor communication between users. These issues highlight the need for an automated and intelligent system.

Based on this analysis, the proposed system is designed to incorporate advanced technologies such as NLP and AI. The system aims to automate project submission, detect duplicate ideas, evaluate project quality, and streamline the approval workflow.

Overall, system analysis provides a clear understanding of the problem do-main and lays the foundation for designing a robust, scalable, and efficient Academic Project Selection System.

3.1 Software Requirement Specification

The implementation of the Academic Project Selection System requires a well-defined set of hardware and software resources to ensure efficient performance, reliability, and scalability. These requirements are essential for both development and deployment phases of the system.

3.1.1 User Requirements

The Academic Project Selection System is designed to provide a comprehensive, user-friendly platform for students, guides, and administrators in academic institutions to manage, validate, and track student project submissions efficiently. From a student's perspective, the system allows secure registration and login, ensuring that each user has a personal account with access controlled according to their role.

Students can submit detailed project proposals that include essential information such as the project title, abstract, keywords, technologies used, methodology, problem statement, objectives, team members, guide name, branch, and semester.

Each submission undergoes AI-powered validation to check for completeness, clarity, and adherence to academic standards, providing students with constructive feed-back to improve the quality of their proposals. In addition, the system incorporates advanced NLP-based similarity detection, which evaluates new submissions against existing projects and calculates similarity scores, helping students avoid duplication and maintain originality in their work.

Guides benefit from a role-specific interface that enables them to review projects, provide detailed feedback, and approve or reject proposals based on evaluation criteria. Students can track these reviews, view feedback, and make necessary revisions, ensuring a clear communication loop between project creators and reviewers.

Administrators have access to advanced management features, including the ability to oversee all users, monitor project submissions, and directly insert or manage pro-jects for testing or bulk data operations.

The system offers an intuitive and responsive dashboard tailored to the user's role: students can view their own submissions and status updates, guides can monitor pro-jects assigned to them, and administrators can access a comprehensive overview

of all projects in the system. The platform supports seamless searching, filtering, and navigation of project records, enhancing usability and efficiency.

Furthermore, the system ensures secure storage of data, protects user privacy, and supports concurrent access by multiple users, making it reliable for institutional deployment.

By combining AI validation, NLP similarity detection, role-based access control, and a responsive user interface, the Academic Project Selection System effectively streamlines the academic project management process, providing a robust, accurate, and user-centred solution.

3.1.2 Software Requirements

Table 3.1.1 Software Requirements

Category	Technologies/Tools
Frontend	React.js
Backend	Python (FastAPI)
Libraries / Packages	Scikit-learn, Pandas, NumPy
NLP & Similarity	Natural Language Processing, Cosine Similarity
Database	PostgreSQL
APIs	REST APIs
Operating System	Windows / Linux

3.1.3 Hardware Requirements

Table 3.1.2 Hardware Requirements

Component	Minimum Requirement
Processor	Intel i3 or equivalent CPU
RAM	4 GB
Storage	20 GB free disk space
System Type	Standard 64-bit computer
Frontend Operations	Rendering interface, handling sessions
Backend Operations	Processing API calls, database interactions
AI/NLP Computations	Basic processing
Cloud Deployment	Compatible
Accessibility	Standard computing devices

3.1.4 Libraries and Tools Used

The development of the Academic Project Selection System relies on various libraries and tools that provide essential functionalities such as web development, data processing, and intelligent analysis. These tools enhance system performance, simplify development, and ensure accurate results.

React is a powerful JavaScript library used for building user interfaces. It enables the development of dynamic and interactive web pages using reusable components. In the Academic Project Selection System, React is used to design the frontend interface, including forms for project submission, dashboards for tracking status, and pages for displaying feedback. React improves user experience by enabling fast rendering and seamless navigation. Its component-based architecture ensures that the system is modular, maintainable, and scalable

Scikit-learn is a widely used machine learning library in Python that provides efficient tools for data analysis and modeling. In this system, it is primarily used for implementing NLP techniques such as TF-IDF vectorization and cosine similarity. Scikit-learn simplifies the process of converting text data into numerical form and comparing different project descriptions. It ensures accurate and efficient similarity detection, which is a core feature of the system.

Pandas is a data manipulation and analysis library used for handling structured data. It provides powerful data structures such as Data Frames, which make it easy to organize, clean, and preprocess data. In this system, Pandas is used to manage project data, perform preprocessing operations, and prepare data for analysis. It plays a crucial role in ensuring data consistency and improving processing efficiency.

NumPy is a fundamental library for numerical computing in Python. It provides support for large, multi-dimensional arrays and mathematical operations. In the Academic Project Selection System, NumPy is used to perform vector calculations required for similarity analysis. It enhances computational efficiency and ensures faster processing of large datasets.

NLP techniques are used to analyse and process textual data. TF-IDF is used to convert text into numerical vectors by assigning weights to words based on their importance. Cosine similarity is used to measure the similarity between these vectors. It

calculates the cosine of the angle between two vectors, providing a similarity score. These techniques are essential for detecting duplicate or similar project ideas and ensuring originality.

PostgreSQL is a powerful and open-source relational database management system. It is used to store all application data in a structured format. It supports complex queries, ensures data integrity, and provides high scalability. PostgreSQL plays a critical role in managing project data, user information, and system outputs efficiently. REST APIs enable communication between the frontend and backend components of the system. They allow data to be sent and received in a structured format, typically JSON. In this system, REST APIs handle operations such as project submission, retrieving project data, performing similarity checks, and displaying evaluation results. This ensures seamless integration between different modules.

The system is designed to be platform-independent and can run on various operating systems, including Windows, Linux, and macOS. This flexibility allows developers and users to work in their preferred environment without compatibility issues.

3.2 Content Diagram of Project

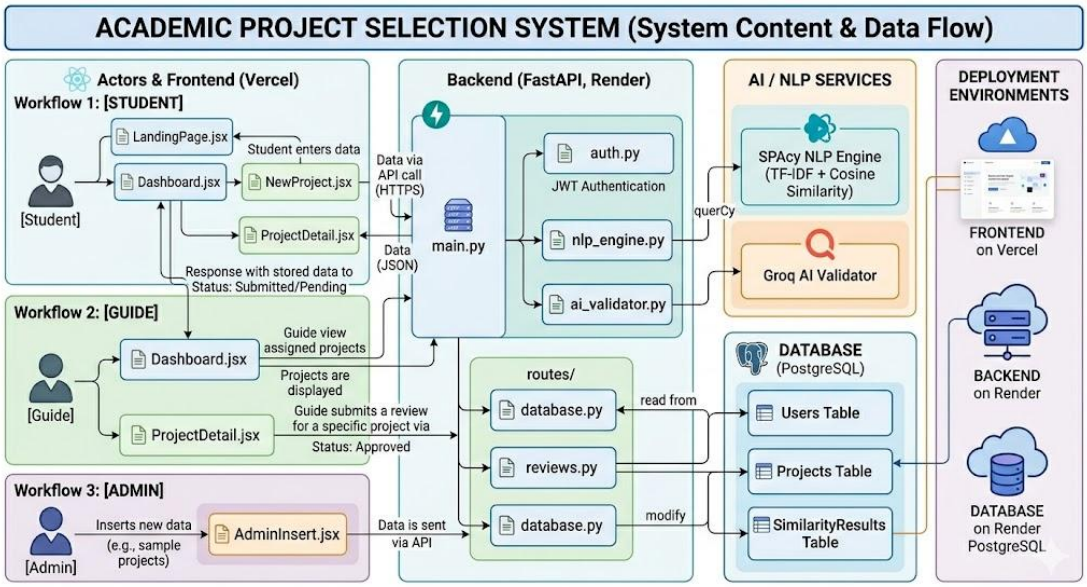


Figure 3.2.1 Top-Level Structure

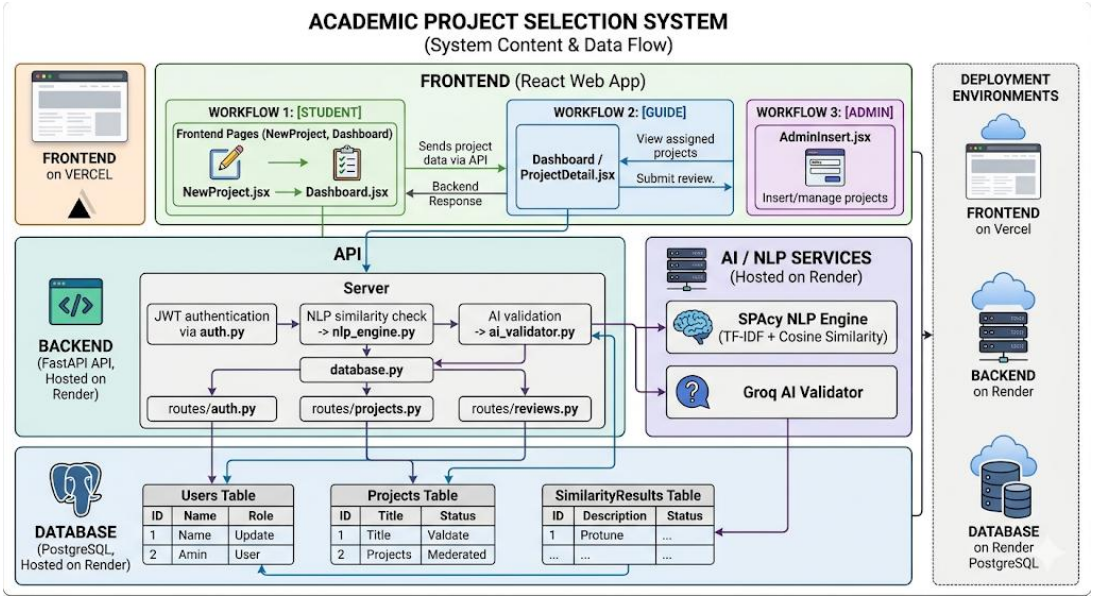


Figure 3.2.2 Flow of Content

3.2.1 Diagram Explanation

The content diagram of the Academic Project Selection System provides a clear overview of the major components, their interactions, and the flow of data throughout the system.

The **frontend** is responsible for all user interactions and serves as the primary interface through which students, guides, and administrators access the system. It includes role-specific dashboards, forms for project submission, pages for viewing project details, and review interfaces. The frontend ensures a responsive and intuitive user experience, allowing users to easily navigate, search, and filter projects based on their role and permissions. All user inputs, such as project submissions or reviews, are sent to the backend through secure API calls.

The **backend** is the core processing unit of the system. It handles all API requests from the frontend, including authentication, project submissions, similarity checks, and review updates. The backend integrates with AI and NLP modules to perform tasks such as validating project quality using Groq AI and checking for duplicate or similar project submissions using spaCy NLP and TF-IDF with cosine similarity. It also enforces role-based access control, ensuring that students, guides, and adminis-

trators can only perform actions allowed by their permissions. The backend communicates directly with the database to store, retrieve, and update all system data efficiently.

The **database**, powered by PostgreSQL, serves as the central repository for all system data. It stores structured information including student, guide, and admin accounts, project submissions, similarity results, and project review history. By organizing data in relational tables with proper keys and constraints, the database ensures data integrity, consistency, and reliability, allowing the system to retrieve information quickly and accurately whenever needed.

The **AI/NLP components** provide intelligent validation and analysis capabilities. The NLP engine evaluates the similarity between new project submissions and existing projects to detect duplicates or highly similar proposals, maintaining the originality of all submissions. The AI validation module analyzes project content for completeness, clarity, and adherence to academic standards, providing constructive feedback to students to improve their proposals. These components enhance the overall quality and credibility of the system.

The **deployment layer** defines where each component is hosted and how users access the system. The frontend is deployed on Vercel, offering a scalable and fast web interface, while the backend and PostgreSQL database are hosted on Render, providing reliable cloud-based processing and storage. Deployment ensures that all components are accessible over the internet, allowing multiple users to interact with the system simultaneously and supporting scalability for handling larger datasets and higher traffic.

Overall, the content diagram illustrates a well-integrated, modular system where the frontend, backend, database, and AI/NLP components work together seamlessly, providing a robust platform for managing academic project submissions and reviews.

4 Design of the Project

4.1 Introduction

The **Academic Project Selection System** is designed to provide a structured, intelligent, and user-friendly platform for managing student academic projects in educational institutions. The design focuses on ensuring that students can submit original projects, guides can evaluate them efficiently, and administrators can oversee the entire process in a streamlined manner.

The system adopts a full-stack, three-tier architecture that separates the frontend, backend, and database layers, promoting modularity, maintainability, and scalability. Additionally, it integrates Artificial Intelligence and Natural Language Processing modules to automatically validate project quality and detect similarities among submissions.

This design ensures that the system is efficient, accurate, and scalable, providing a seamless user experience for all roles—students, guides, and administrators. By combining modern web technologies, AI/NLP capabilities, and cloud deployment, the system addresses the challenges of project duplication, quality validation, and centralized project management effectively.

4.2 DFD / ER Diagrams

A **Data Flow Diagram (DFD)** is a graphical representation that shows how data moves through a system. It illustrates the flow of information between external entities, processes, and data stores.

DFDs help in understanding system functionality, data processing, and interactions in a simple and structured way.

4.2.1 DFD Level 0

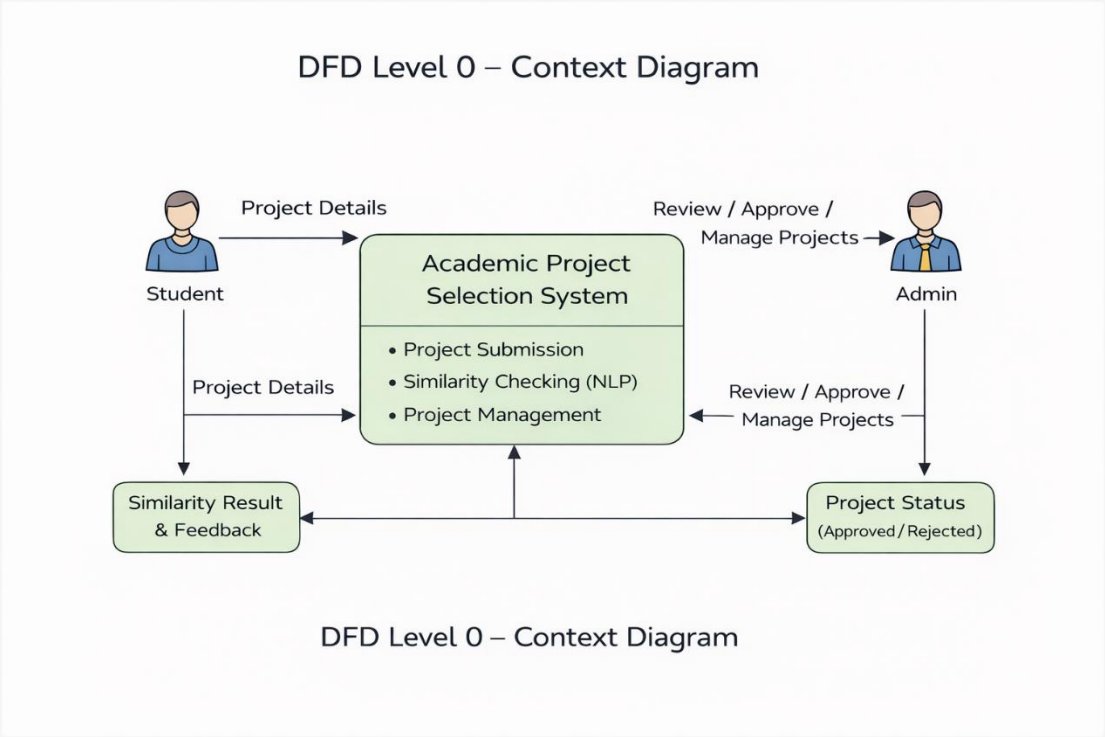


Figure 4.2.1 DFD Level 0

DFD Level 0, also known as the **Context Diagram**, represents the system at the highest level of abstraction. It shows the entire system as a **single process** and highlights its interaction with external entities.

This level does not provide internal details of the system; instead, it focuses on the overall data exchange between the system and users.

The **Academic Project Selection System** is represented as one main process. The system interacts with two primary external entities:

- **Student**
- **Admin**

Data Flow Description:

- The **student** submits project details such as title, abstract, description, and keywords to the system.
- The system processes the input and returns **similarity results and feedback** to the student.
- The **Admin** receives the project for review from the system.
- The admin sends back the **approval or rejection decision**.
- The system updates and stores the final project status.

This level provides a **bird's-eye view** of the system and helps in understanding the overall interaction without going into technical details.

4.2.2 DFD Level 1

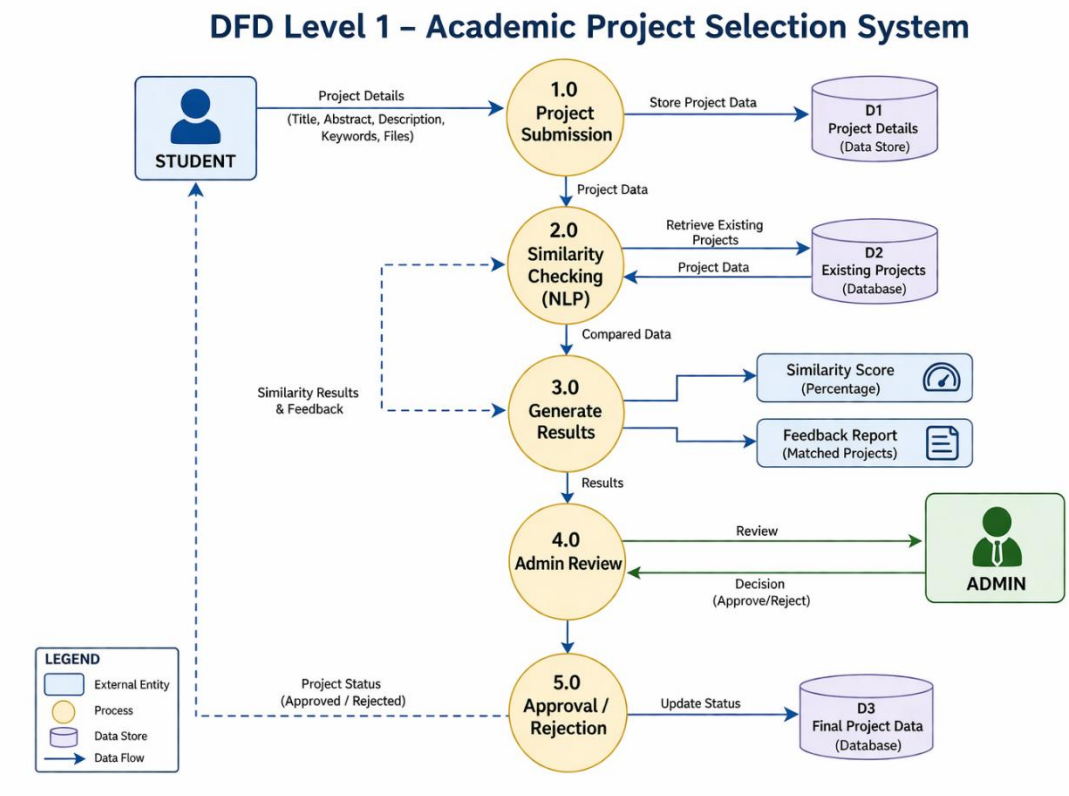


Figure 4.2.2 DFD Level 1

DFD Level 1 provides a **detailed decomposition** of the main process shown in Level 0. It breaks the system into multiple sub-processes and shows how data flows between them.

This level gives a clearer understanding of how the system works internally by showing step-by-step data processing.

In this project, the system is divided into the following processes:

1. Project Submission (Process 1.0)

- The student enters project details including title, abstract, description, and keywords.
- The system stores this information in the **Project Details Data Store**.

2. Similarity Checking (Process 2.0)

- The system retrieves existing project data from the database.
- It performs **NLP-based similarity analysis** to compare the new project with previous submissions.

3. Generate Results (Process 3.0)

- The system generates:
    - **Similarity Score (percentage)**
    - **Matched project details**
    - **Feedback report**
- These results are sent back to the student for review.

4. Admin Review (Process 4.0)

- The admin reviews the submitted project along with similarity results.
- Based on evaluation, the admin decides whether to approve or reject the project.

5. Approval / Rejection (Process 5.0)

- The final decision is recorded.
- The system updates the **Final Project Database** with the project status (Approved/Rejected).
- The updated status is communicated to the student.

Data Stores Used:

- **D1: Project Details** – Stores submitted project information
- **D2: Existing Projects Database** – Contains previous project records for comparison
- **D3: Final Project Records** – Stores approved/rejected project data

DFD Level 1 provides a **complete internal workflow** of the system, showing how data is processed at each stage and how different components interact with each other.

4.2.3 ER Diagram

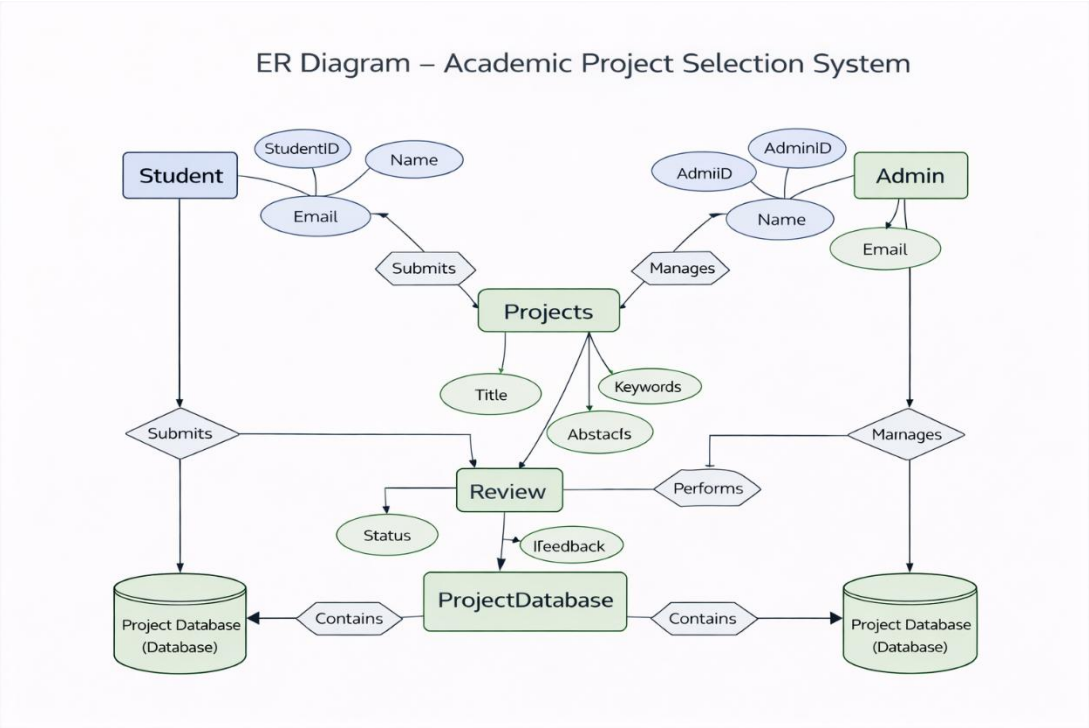


Figure 4.2.3 ER Diagram

Entity Relationship (ER) Diagram

An **Entity Relationship (ER) Diagram** is a graphical representation used to model the structure of a database. It shows how different **entities (tables)** are related to each other within a system. ER diagrams help in designing and organizing data efficiently before implementing it in a database.

The main components of an ER diagram include:

- **Entities** – Real-world objects or concepts (e.g., Student, Project, Admin)
- **Attributes** – Properties of entities (e.g., Name, Email, Project Title)
- **Relationships** – Connections between entities (e.g., Student submits Project)
- **Primary Key** – Unique identifier for each entity
- **Foreign Key** – Attribute used to link entities

In this project, the ER diagram represents the database structure used for managing academic project submissions, similarity checking, and approval processes.

Entities and Their Description

1. Student

This entity stores information about the students who submit projects.

Attributes:

- Student_ID (Primary Key)
- Name
- Email
- Department
- Academic_Year

2. Project

This entity contains details of the submitted projects.

Attributes:

- Project_ID (Primary Key)
- Title
- Abstract
- Description
- Keywords
- Similarity_Score
- Status (Pending/Approved/Rejected)

3. Admin

This entity represents the administrator who reviews projects.

Attributes:

- Admin_ID (Primary Key)
- Name
- Email

4. Database / Records

This represents stored project data used for similarity comparison.

Attributes:

- Record_ID (Primary Key)
- Project_ID (Foreign Key)
- Stored_Data

Relationships

- A **Student submits Project**
  → One student can submit one or more projects (One-to-Many relationship)
- A **Project is reviewed by Admin**
  → One admin can review multiple projects (One-to-Many relationship)
- A **Project is compared with Existing Records**
  → Used for similarity checking (Many-to-One relationship)

Conclusion

The ER diagram clearly defines how data is structured and related in the system. It ensures efficient database design, minimizes redundancy, and supports smooth implementation of features like project submission, similarity checking, and admin approval.

4.3 Data Dictionary

A **Data Dictionary** is a structured repository that defines all the data elements used in the system. It provides detailed information about each attribute, including its name, data type, size, and description. The data dictionary ensures consistency, accuracy, and clarity in database design and helps developers and users understand how data is stored and managed.

Table 4.3.1 Users Table

Column	Type	Description
id	Integer (PK)	Unique identifier
full_name	String (150)	Full name of user
email	String (200)	Unique email address
hashed_password	String (300)	Encrypted password
department	String (150)	Department of the user

Column	Type	Description
role	String (50)	Role: student, guide, admin
is_active	Boolean	Account active status
created_at	DateTime	Account creation timestamp

- The id acts as the Primary Key (PK) ensuring each user is uniquely identified.
- The email field is unique and used for authentication.
- The hashed_password ensures secure storage of user credentials.
- The role attribute enables role-based access control, allowing different functionalities for students, guides, and admins.
- The is_active field helps in managing user access (active/inactive accounts).
- The created_at field is useful for tracking user registration history.

Table 4.3.2 Projects Table

Column	Type	Description
id	Integer (PK)	Unique project ID
owner_id	Integer (FK)	References users.id
title	String (300)	Project title
abstract	Text	Project summary
keywords	Text	Project keywords
technologies	Text	Technologies used
methodology	Text	Project methodology

Column	Type	Description
problem_statement	Text	Project problem
objectives	Text	Project objectives
team_members	Text	Names of team members
guide_name	String (150)	Assigned guide
branch	String (50)	Student branch
semester	String (20)	Student semester
status	String (30)	approved, rejected
overall_similarity	Float	NLP similarity score
created_at	DateTime	Submission timestamp

Table 4.3.3 Similarity Results Table

Column	Type	Description
id	Integer (PK)	Unique result ID
project_id	Integer (FK)	Source project
compared_with_id	Integer (FK)	Compared project
overall_score	Float	Overall similarity percentage
field scores	JSON	Field-wise similarity scores

Column	Type	Description
created_at	DateTime	Comparison timestamp

Table 4.3.4 Reviews Table

Column	Type	Description
id	Integer (PK)	Unique review ID
project_id	Integer (FK)	Project being reviewed
reviewer_id	Integer (FK)	Guide performing the review
comments	Text	Feedback from guide
status	String (30)	status: approved, rejected
created_at	DateTime	Review submission timestamp

5 Implementation and Results

The implementation phase is one of the most important stages in the development of the Academic Project Selection System. It involves transforming the system design into a functional and operational application using appropriate technologies and tools. This phase ensures that all planned features are correctly developed, integrated, and tested to meet the system requirements.

The Academic Project Selection System is implemented as a web-based application that provides a seamless interface for students, guides, and administrators. The system follows a modular architecture, where each component is designed to perform specific tasks such as project submission, similarity detection, quality evaluation, and approval management. This modular approach improves maintainability, scalability, and ease of future enhancements.

The frontend of the system is developed using React, which provides an interactive and responsive user interface. It allows users to easily submit project proposals, view feedback, and track project status. The backend is implemented using FastAPI, which handles API requests, processes data, and manages communication between different components of the system. FastAPI is chosen for its high performance and ability to handle asynchronous operations efficiently.

The system uses PostgreSQL as the database to store all relevant information, including user details, project submissions, similarity scores, and evaluation results. Proper database design ensures data integrity, consistency, and efficient retrieval of information. A key feature of the system is the integration of Natural Language Processing (NLP) techniques for similarity detection. The system uses TF-IDF to convert textual project descriptions into numerical vectors and cosine similarity to measure the similarity between projects. This helps in identifying duplicate or closely related project ideas, ensuring originality and innovation.

In addition to NLP, Artificial Intelligence (AI) techniques are used to evaluate the quality of project proposals. The AI module analyses factors such as clarity, relevance, and completeness of the project description. Based on this analysis, the system pro-

vides feedback to students, helping them improve their submissions before final approval.

The implementation also includes role-based access control, where different users (students, guides, and administrators) have specific permissions and functionalities. Students can submit and track projects, guides can review and provide feedback, and administrators can approve or reject projects. The system is tested at different stages to ensure reliability and performance. Unit testing, integration testing, and user acceptance testing are conducted to verify that all modules function correctly. Error handling and validation mechanisms are implemented to prevent invalid inputs and ensure smooth system operation.

Overall, the implementation of the Academic Project Selection System provides an efficient, automated, and intelligent solution for managing academic projects. It reduces manual effort, improves accuracy, enhances transparency, and supports better decision-making in the project selection process.

5.1 Method of Implementation

The implementation of the Academic Project Selection System follows a structured and systematic approach to ensure efficiency, scalability, and user-friendliness. The development process began with requirement analysis, where functional needs such as project submission, review, AI-based validation, similarity detection, and role-based dashboards were identified, along with non-functional requirements including performance, security, and usability.

Based on these requirements, a three-tier system architecture was designed, consisting of the frontend, backend, and database layers, with additional AI and NLP components for intelligent project evaluation.

The frontend was implemented using React, Tailwind CSS, and Vite, creating responsive, interactive, and modular interfaces for students, guides, and administrators. Role-based dashboards, submission forms, and project detail views were developed as reusable components.

React Router was used to manage navigation, while Axios handled secure communication with the backend via REST APIs, automatically attaching JWT tokens for authentication.

The backend, built using FastAPI and Python 3.11, handled business logic, API requests, authentication, AI/NLP validation, and database interactions. The system used JWT-based authentication for secure login and bcrypt for password hashing.

The NLP similarity detection algorithm employed TF-IDF vectorization and cosine similarity to identify duplicate or similar projects, while the AI validation module used the Groq API to assess project completeness, clarity, and academic quality. These modules were integrated into the backend so that every submission was automatically analyzed before storage.

The database, implemented in PostgreSQL, was designed to store users, projects, reviews, and similarity results in a structured manner. SQLAlchemy ORM and asyncpg were used for asynchronous operations, ensuring fast and reliable data access. All database models were clearly defined, enforcing integrity and supporting relational dependencies between projects, users, and reviews.

The implementation process also included rigorous testing, such as unit testing for individual components, integration testing for frontend-backend communication, and user acceptance testing to verify that the system met all functional and non-functional requirements.

Finally, the system was deployed on cloud platforms, with the frontend hosted on Vercel and the backend and database on Render, ensuring scalability, remote accessibility, and reliable performance. Environment variables were securely configured for API keys, database connections, and JWT secrets, completing the end-to-end deployment.

Overall, this method of implementation emphasizes a modular, maintainable, and scalable design, integrating modern web technologies and intelligent algorithms to deliver a secure, efficient, and user-friendly academic project management system.

5.1.1 Implementation of Algorithms

The Academic Project Selection System incorporates intelligent algorithms to ensure the originality and quality of student project submissions. The core algorithm for similarity detection uses Natural Language Processing (NLP) techniques, particularly TF-IDF vectorization and cosine similarity, to compare new submissions with existing projects in the database.

When a student submits a project, the system first preprocesses the text by removing punctuation, stop words, and special characters, and tokenizes it into individual words. Each project's textual data, including the title, abstract, keywords, methodology, and technologies, is then converted into numerical vectors using TF-IDF, which assigns weights to words based on their frequency in the project relative to all other submissions.

Cosine similarity is subsequently calculated between these vectors to generate a similarity score ranging from 0 to 100%, with higher scores indicating greater similarity. Projects that exceed a predefined threshold are flagged as potentially duplicate, and detailed field-wise similarity results are stored in the database for further reference.

In addition to similarity detection, the system integrates an AI-based project validation algorithm using the Groq API. This algorithm evaluates the quality, completeness, and clarity of each submission.

The backend compiles all project fields, formats them appropriately, and sends them to the AI model for analysis. The AI assesses logical consistency, academic standards, and completeness, providing a quality score along with constructive feedback. Submissions scoring below the threshold receive detailed guidance for improvement, which is communicated back to the student through the frontend interface.

Together, the NLP similarity detection and AI validation algorithms form a comprehensive evaluation framework. Upon project submission, the backend automatically triggers both algorithms: the AI validation checks the content quality, while the NLP engine verifies originality.

The results, including similarity scores, field-specific comparisons, and AI feedback, are stored in the PostgreSQL database and displayed to both students and guides. This integrated approach ensures that submitted projects are unique, high-quality, and academically sound, while also reducing manual evaluation effort and enhancing the efficiency of project management within the academic institution.

NLP-Based Similarity Detection Algorithm: The similarity detection algorithm ensures that submitted projects are original and not duplicates of existing projects in the system. It uses Natural Language Processing (NLP) techniques, particularly TF-IDF (Term Frequency–Inverse Document Frequency) and cosine similarity, to quantify the similarity between project documents.

Text Preprocessing: Project fields such as title, abstract, keywords, methodology, and technologies are extracted. Text is cleaned by removing punctuation, stop words, and special characters. Tokenization is performed to split text into individual words. Words are converted to lowercase for uniformity.

Vectorization using TF-IDF: Each project's text is transformed into a numerical vector using the TF-IDF approach. TF-IDF assigns weights to words based on their frequency in a project relative to all projects, emphasizing important words and reducing the impact of common terms.

Formula:

$$\text{TF-IDF}(t,d) = TF(t,d) \times \log \frac{N}{DF(t)}$$

Where:

- $TF(t,d)$= Frequency of term $t$in document d
- $DF(t)$= Number of documents containing term t
- N= Total number of documents

Cosine Similarity Calculation: Cosine similarity is used to measure the angle between two TF-IDF vectors, providing a score between 0 and 1 (or 0–100%). Higher values indicate greater similarity between two projects.

Formula:

$$\text{Cosine Similarity} = \frac{\vec{A} \cdot \vec{B}}{\| \vec{A} \| \| \vec{B} \|}$$

Where:

- $\vec{A}$and $\vec{B}$are TF-IDF vectors of two project submissions

Similarity Threshold: Projects with similarity scores above a predefined threshold (e.g., 70%) are flagged as potentially duplicate. Detailed field-wise similarity scores are stored in the database for reference.

Integration: Implemented using Python's scikit-learn and spaCy libraries. Triggered automatically when a new project is submitted or updated.

AI-Based Project Validation Algorithm: The AI validation algorithm evaluates the quality, completeness, and clarity of project submissions using a Large Language Model (LLM) through the Groq API.

Data Preparation: All project fields (title, abstract, methodology, objectives, problem statement, technologies, team members) are compiled into a structured input format. The input text is cleaned and formatted for API submission.

AI Request to Groq API: The backend sends the structured project data to Groq's AI model using an API call. The AI model analyzes the submission for Completeness of information, Logical consistency in objectives and methodology, Academic standard of language and structure and Suggestions for improvements

AI Response Handling: The AI returns a quality score (e.g., 0–100%) and feedback comments. Scores below a defined threshold indicate areas that require improvement. Feedback is sent to the student through the frontend interface for revisions. Integration Implemented as a separate ai_validator.py module in the backend. Invoked automatically during project submission or modification.

The Academic Project Selection System is designed as an advanced, web-based solution to overcome the limitations of traditional systems. It provides a centralized

platform where students, guides, and administrators can interact efficiently and manage project-related activities.

The system integrates modern technologies such as Natural Language Processing (NLP) and Artificial Intelligence (AI) to automate and enhance the project selection process.

NLP techniques are used to analyse project descriptions and detect similarities with previously submitted projects, ensuring originality. AI-based models are used to evaluate the quality of project proposals and provide constructive feedback.

The system supports multiple user roles, each with specific functionalities. Students can submit project ideas and track their status. Guides can review submissions and provide feedback, while administrators can approve or reject projects. This structured workflow ensures proper coordination and accountability at each stage.

By automating repetitive tasks and providing intelligent insights, the system improves efficiency, reduces manual workload, and enhances the overall quality of academic projects.

The system follows a well-defined and structured workflow to ensure efficient processing of project submissions. Initially, students submit their project proposals through the web interface. The system then processes the input data and performs similarity analysis using NLP techniques.

Once logged in, a student can submit a new project by filling in details such as title, abstract, keywords, technologies, methodology, and problem statement. After submission, the backend processes the input data and performs text preprocessing, which includes cleaning, tokenization, and normalization of the text. The processed data is then converted into numerical form using TF-IDF vectorization.

Next, the system performs similarity analysis by comparing the submitted project with existing projects stored in the database using cosine similarity. A similarity score is generated, and if the score exceeds a predefined threshold, the project is flagged as potentially duplicate.

Alongside this, the system sends the project data to the AI validation module, which evaluates the quality, completeness, and clarity of the project and returns feedback along with a validation score.

After processing, the project along with its similarity results and AI feedback is stored in the PostgreSQL database. The student can view the results on their dashboard, including similarity percentage and AI suggestions.

Guides can access submitted projects, review the details, provide feedback, and approve or reject the project. Administrators have full control over the system, including managing users, projects, and overall system operations.

If the project passes the similarity check, it is further evaluated by the AI module, which analyzes various aspects such as clarity, relevance, and completeness. Based on this evaluation, the system generates feedback to help students improve their proposals.The project is then forwarded to the guide for review. The guide can either approve the project, request modifications, or reject it based on its quality and feasibility.

Finally, the administrator reviews the project and makes the final decision regarding approval. This structured workflow ensures transparency, accountability, and efficiency throughout the process.

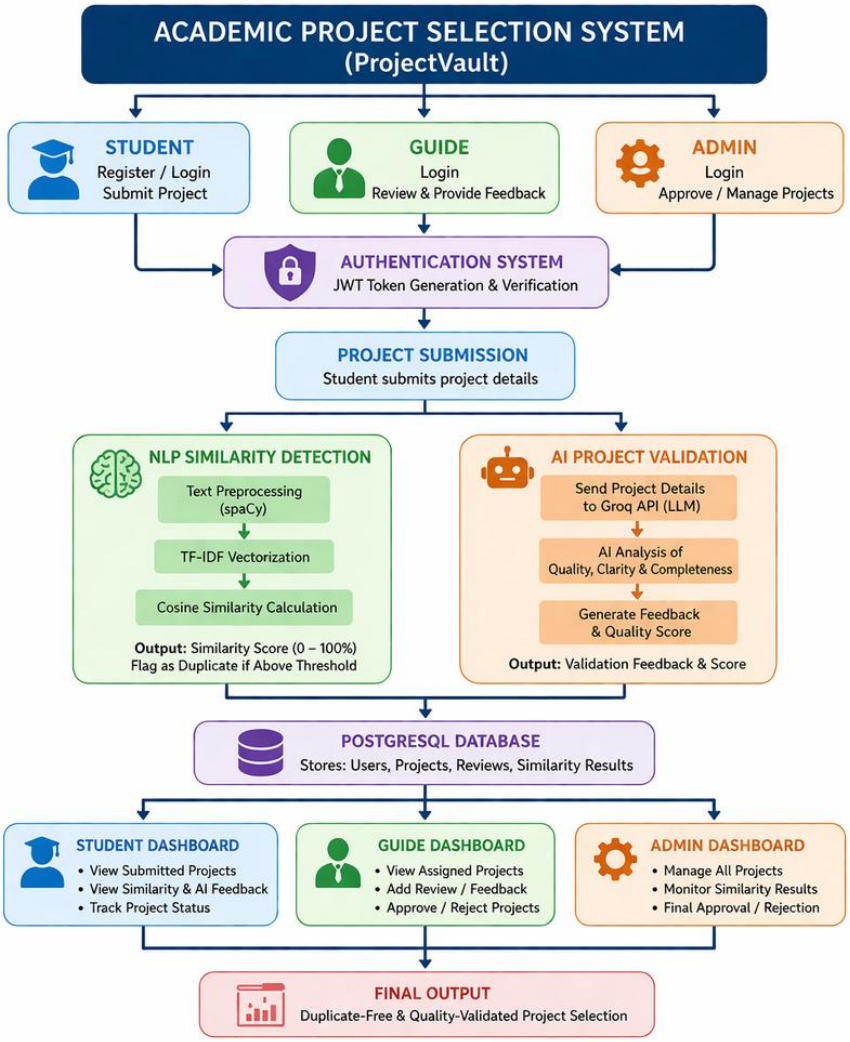


Figure 5.1.1 Work Flow

Automation plays a crucial role in modernizing academic systems. With the increasing number of students and project submissions, manual processes are no longer sufficient. An automated system can handle large volumes of data efficiently and reduce the burden on faculty members.

Automation ensures consistency in evaluation by applying predefined criteria and algorithms. It reduces human errors and speeds up the overall process. Additionally, it enables real-time tracking of project status, improving transparency and communication.

By incorporating automation, institutions can promote innovation by ensuring that students work on unique and meaningful projects. It also allows faculty members to focus on guidance and mentoring rather than administrative tasks.

The Academic Project Selection System uses advanced similarity detection techniques to compare newly submitted project ideas with existing ones in the database. This is essential to ensure originality and avoid duplication of project topics among students.

One of the key techniques used is Term Frequency–Inverse Document Frequency (TF-IDF). This method converts textual data into numerical vectors by assigning weights to words based on their importance in a document.

Words that occur frequently in a specific document but are rare across other documents are given higher importance. This helps in representing the project description in a meaningful mathematical form.

After vectorization, Cosine Similarity is applied to measure the similarity between two project descriptions. It calculates the cosine of the angle between two vectors in a multi-dimensional space. A value closer to 1 indicates high similarity, while a value closer to 0 indicates low similarity. This method is efficient and widely used in text comparison tasks.

By combining TF-IDF and cosine similarity, the system can accurately identify duplicate or closely related project ideas. This not only prevents repetition but also encourages students to come up with unique and innovative project proposals.

The scope of the Academic Project Selection System includes the development of a comprehensive platform to manage the entire lifecycle of academic projects, from submission to approval. The system is intended for use in colleges and universities to streamline project management processes and improve efficiency.

The current system focuses on project submission, similarity detection, quality evaluation, and role-based access for students, guides, and administrators. It ensures proper tracking, transparency, and structured workflow within the institution.

In the future, the system can be extended with additional advanced features such as:

- Plagiarism detection to ensure complete originality of project content
- Recommendation systems to suggest relevant project topics to students
- Mobile application support for better accessibility and user experience
- Advanced AI models for more accurate evaluation and intelligent feedback
- Integration with academic databases for better data management

The system is scalable and flexible, allowing it to adapt to different institutional requirements and technological advancements.

5.1.2 Methodology of the Project

The methodology of the proposed system defines a systematic approach for managing academic project submissions using advanced technologies such as Natural Language Processing (NLP) and Artificial Intelligence (AI).

The goal of this methodology is to automate the process of project evaluation, detect duplication, and enhance the overall quality of project ideas.The system follows a step-by-step process starting from project submission to final approval, ensuring accuracy, transparency, and efficiency.

5.1.3 Proposed Methodology

The proposed methodology is divided into multiple stages, each responsible for a specific function within the system.

Stage 1: Project Submission:

Students submit their project proposals through the system interface. The submission includes: Project title, Description and Domain or category. This information is stored in the database for further processing.

Stage 2: Data Preprocessing (NLP):

Once the project is submitted, the NLP module processes the textual data to make it suitable for analysis. This includes: Tokenization (breaking text into words), Removal of stop words (e.g., “is”, “the”, “and”), Stemming or lemmatization and Cleaning and normalization of text. This step ensures that only meaningful and relevant data is used.

Stage 3: Feature Extraction (TF-IDF)

The cleaned text is converted into numerical form using TF-IDF. This technique: Assigns importance to words based on frequency, reduces the impact of common words, and represents the project as a vector. This transformation is necessary for similarity comparison.

Stage 4: Similarity Detection (Cosine Similarity)

The system compares the submitted project with existing projects using cosine similarity. If similarity score is high → project is considered duplicate and If similarity score is low → project is considered unique. This helps in maintaining originality and avoiding repeated ideas.

Stage 5: AI-Based Quality Evaluation

After passing similarity checks, the AI module evaluates the quality of the project. The evaluation is based on: Clarity of idea. Structure of content, relevance to domain and Completeness of description. The system generates automated feedback suggesting improvements.

Stage 6: Guide Review

The faculty guide reviews the project along with AI-generated feedback. The guide can: Accept the project, suggest modifications, provide additional feedback. This ensures human validation in the system.

Stage 7: Admin Approval

The admin performs the final verification and decision-making. The admin can: Add the previous project. Delete the project. This step ensures proper control and management.

Stage 8: Result Generation

After approval, the system updates the project status and displays: Final decision (Approved/Rejected). Feedback and suggestions. Students can view the results through the system interface.

- Ensures originality of projects
- Reduces manual workload
- Improves project quality
- Provides automated feedback
- Saves time and increases efficiency

The methodology provides a structured and intelligent approach to project management by integrating NLP and AI technologies. It ensures that project submissions are unique, well-evaluated, and efficiently managed, making it highly beneficial for academic institutions.

5.1.4 Approach of the Project

The approach of the proposed Academic Project Selection System is designed to provide an intelligent, efficient, and automated solution for managing academic project submissions and evaluations.

It integrates modern technologies such as Natural Language Processing (NLP) and Artificial Intelligence (AI) to overcome the limitations of traditional manual systems. The system follows a hybrid approach that combines automated processing with human decision-making to ensure both accuracy and reliability.

Initially, students submit their project proposals through a user-friendly interface, after which the system processes the textual data using NLP techniques such as tokenization, stop-word removal, and text normalization.

The processed data is then converted into numerical form using TF-IDF, enabling effective comparison with existing projects. Cosine similarity is applied to detect duplicate or closely related project ideas, ensuring originality and preventing redundancy.

Once the project passes the similarity check, it is evaluated by the AI module, which analyses the quality of the content based on factors such as clarity, structure, and relevance, and generates useful feedback for improvement.

The system also incorporates a layered architecture approach, consisting of a frontend for user interaction, a backend for processing and communication, an intelligent analysis layer for NLP and AI operations, and a database layer for secure data storage.

In addition to automated evaluation, the system includes human involvement where faculty guides review the projects and provide suggestions, followed by final approval or rejection by the admin.

This combination of automated and manual processes ensures a balanced and transparent decision-making system. Overall, the approach focuses on reducing manual workload, improving efficiency, enhancing project quality, and ensuring a streamlined and reliable project management process in academic institutions.

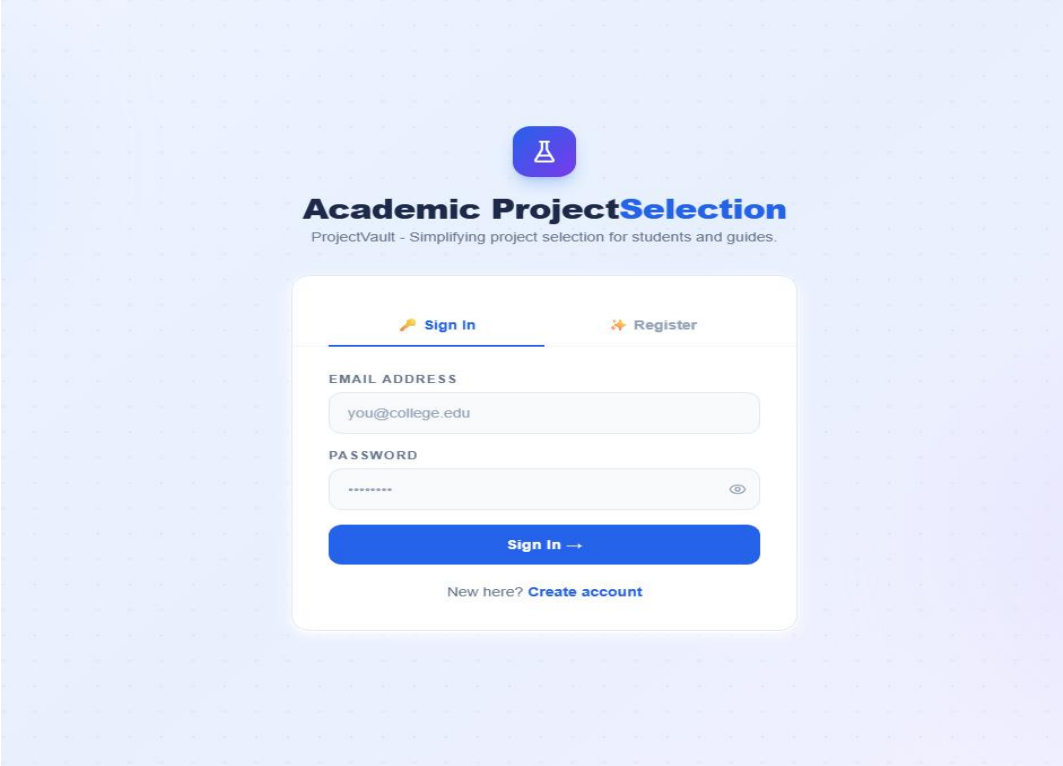
The proposed system follows a structured and transparent workflow for decision making. Initially, the student submits a project proposal, which is then processed by the NLP module for similarity detection.

If the project passes the similarity check, it is forwarded to the AI module for quality evaluation. After automated processing, the project is sent to the guide for review. The guide can approve, reject, or request modifications.

Finally, the administrator reviews the project and makes the final decision. The status of the project is updated in the system, and the student is notified. This workflow ensures proper validation at each stage and improves accountability.

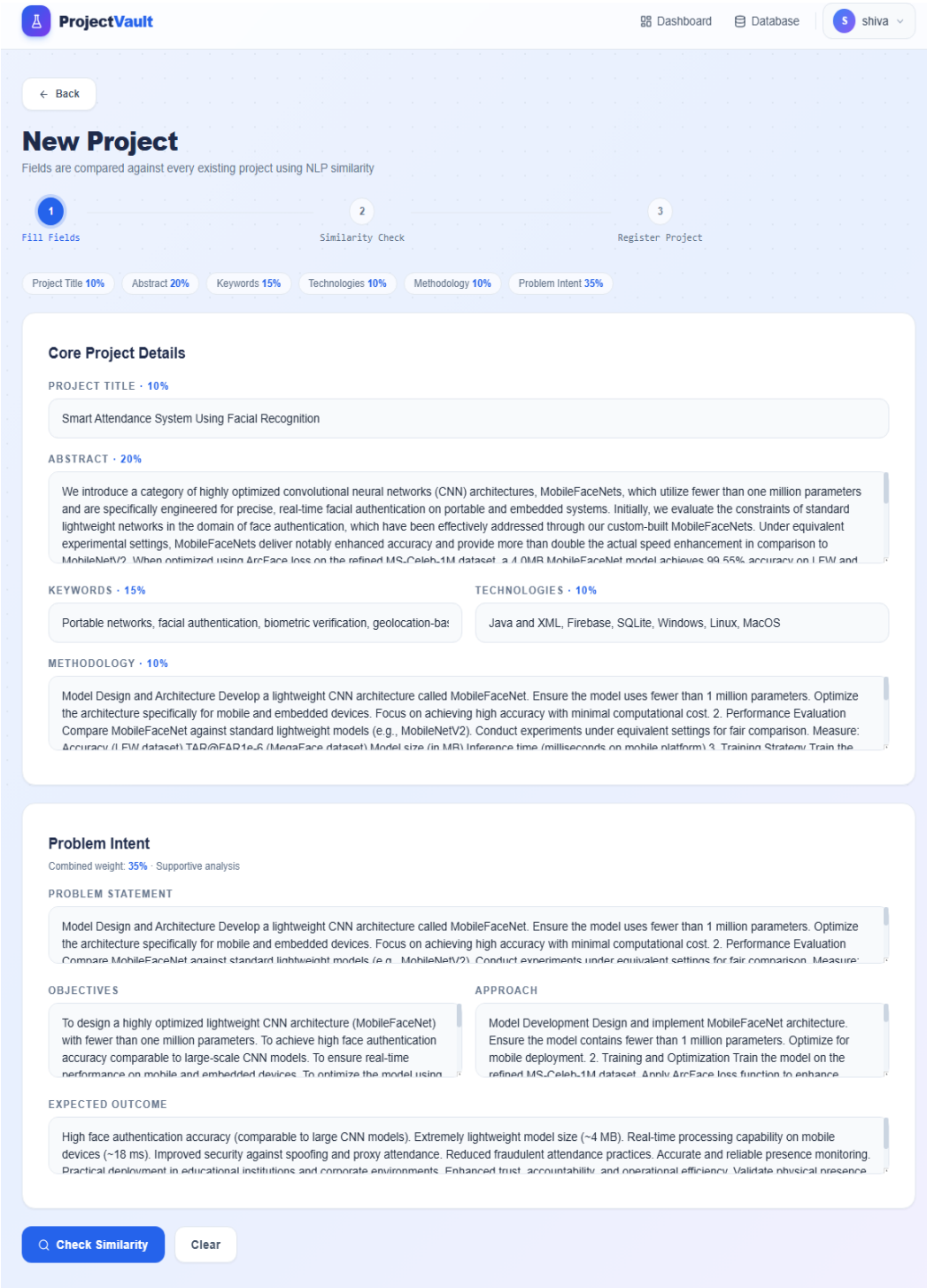
5.1.5 Output Screens

Login Page



The above screen represents the login page of the Academic Project Selection System (Project Vault). It provides a simple and user-friendly interface where users can securely sign in using their email address and password. New users can also create an account using the registration option. This page acts as the entry point to access the system's features such as project submission, similarity checking, and dashboard management.

New Project Submission Page



The above screen shows the New Project Submission page is a key component of the Academic Project Selection System, designed to allow users to enter and manage complete project details in a structured manner. This page provides a user-friendly interface where students can input essential information such as project title, abstract, keywords, technologies used, and methodology. In addition, it includes dedicated sec-

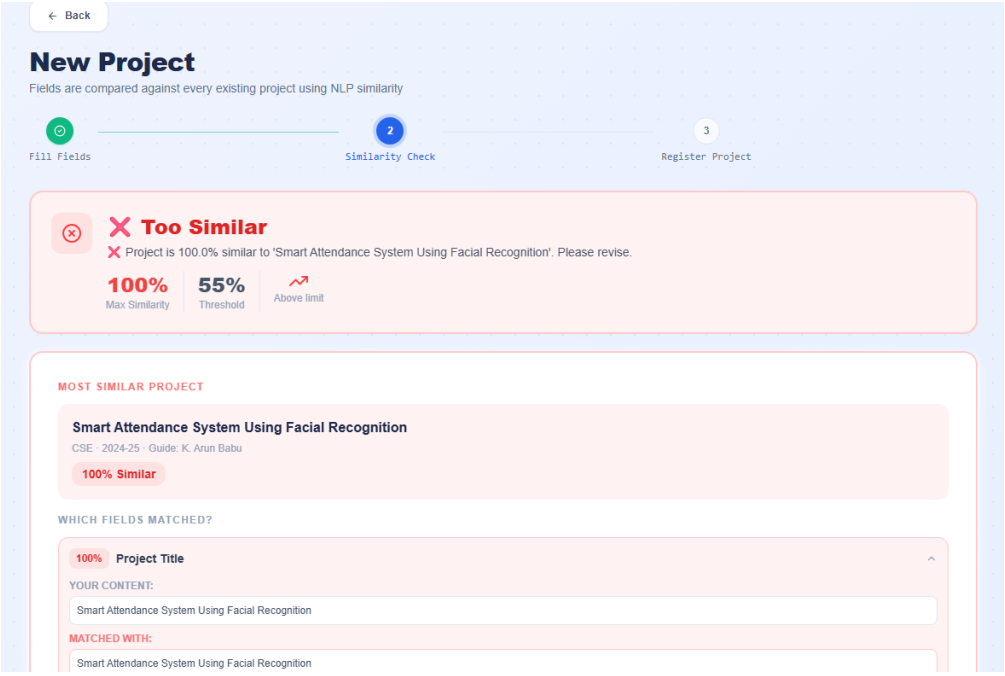
tions for problem statement, objectives, approach, and expected outcomes, ensuring that all aspects of the project are clearly defined and well-documented.

The system follows a step-by-step process, guiding users through stages such as filling in project details, performing similarity checks, and finally registering the project. Each field is assigned a specific weightage, indicating its importance in the overall evaluation process. This structured approach helps maintain consistency and completeness in project submissions.

Once the user enters the details, the system enables a similarity check feature that analyzes the content using NLP techniques. This helps in identifying duplicate or closely related project ideas by comparing the submission with existing records in the database. Based on the results, users can review and refine their content before final submission.

Overall, the New Project Submission page ensures that project entries are accurate, comprehensive, and original. It enhances the quality of submissions while reducing redundancy, making the project selection process more efficient and reliable for both students and evaluators.

Similarity Check Result Page



100% Abstract

YOUR CONTENT:

We introduce a category of highly optimized convolutional neural networks (CNN) architectures, MobileFaceNets, which utilize fewer than one million parameters and are specifically engineered for precise, real-time facial authentication on portable and embedded systems. Initially, we evaluate the constraints of standard lightweight networks in the domain of face authentication, which have been effectively addressed through our custom-built MobileFaceNets. Under equivalent experimental settings, MobileFaceNets deliver notably enhanced accuracy and provide more than double the actual speed enhancement in comparison to MobileNetV2. When optimized using ArcFace loss on the refined MS-Celeb-1M dataset, a 4.0MB MobileFaceNet model achieves 99.55% accuracy on LFW and 92.59% TAR@FAR1e-6 on MegaFace, exhibiting performance similar to large-scale CNN frameworks that are hundreds of megabytes in size. The most efficient MobileFaceNet variant demonstrates an actual processing time of 18 milliseconds on a mobile platform, underscoring exceptional real-time effectiveness. To bolster security and authenticity, we incorporate geolocation-based validation in conjunction with facial recognition. By leveraging GPS and the Fused Location Provider API, the system ensures attendance logging occurs exclusively when the individual is present within a specified geographic zone. This location-aware authentication mitigates fraudulent attendance practices, enhances trustworthiness, and guarantees precise presence monitoring in academic institutions and corporate environments.

MATCHED WITH:

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100% Keywords

YOUR CONTENT:

Portable networks, facial authentication, biometric verification, geolocation-based validation, deep learning, convolutional neural networks.

MATCHED WITH:

Portable networks, facial authentication, biometric verification, geolocation-based validation, deep learning, convolutional neural networks.

100% Technologies

YOUR CONTENT:

Java and XML, Firebase, SQLite, Windows, Linux, MacOS

MATCHED WITH:

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100% Methodology

YOUR CONTENT:

Model Design and Architecture Develop a lightweight CNN architecture called MobileFaceNet. Ensure the model uses fewer than 1 million parameters. Optimize the architecture specifically for mobile and embedded devices. Focus on achieving high accuracy with minimal computational cost. 2. Performance Evaluation Compare MobileFaceNet against standard lightweight models (e.g., MobileNetV2). Conduct experiments under equivalent settings for fair comparison. Measure: Accuracy (LFW dataset) TAR@FAR1e-6 (MegaFace dataset) Model size (in MB) Inference time (milliseconds on mobile platform) 3. Training Strategy Train the model using the refined MS-Celeb-1M dataset. Apply ArcFace loss function to enhance discriminative power. Optimize the model for face authentication tasks. 4. Real-Time Optimization Deploy and test the model on mobile hardware. Measure actual processing time (18 ms reported). Ensure real-time authentication capability. 5. Geolocation-Based Validation Integration Integrate facial recognition with geolocation verification. Use: GPS Fused Location Provider API Define a geographic boundary (geofencing). Allow attendance logging only when: Face authentication is successful, AND User is physically within the authorized zone. 6. Security Enhancement Combine biometric verification + location validation. Prevent proxy attendance and fraudulent logging. Ensure accurate presence monitoring in institutions and corporate settings.

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100% Problem Statement

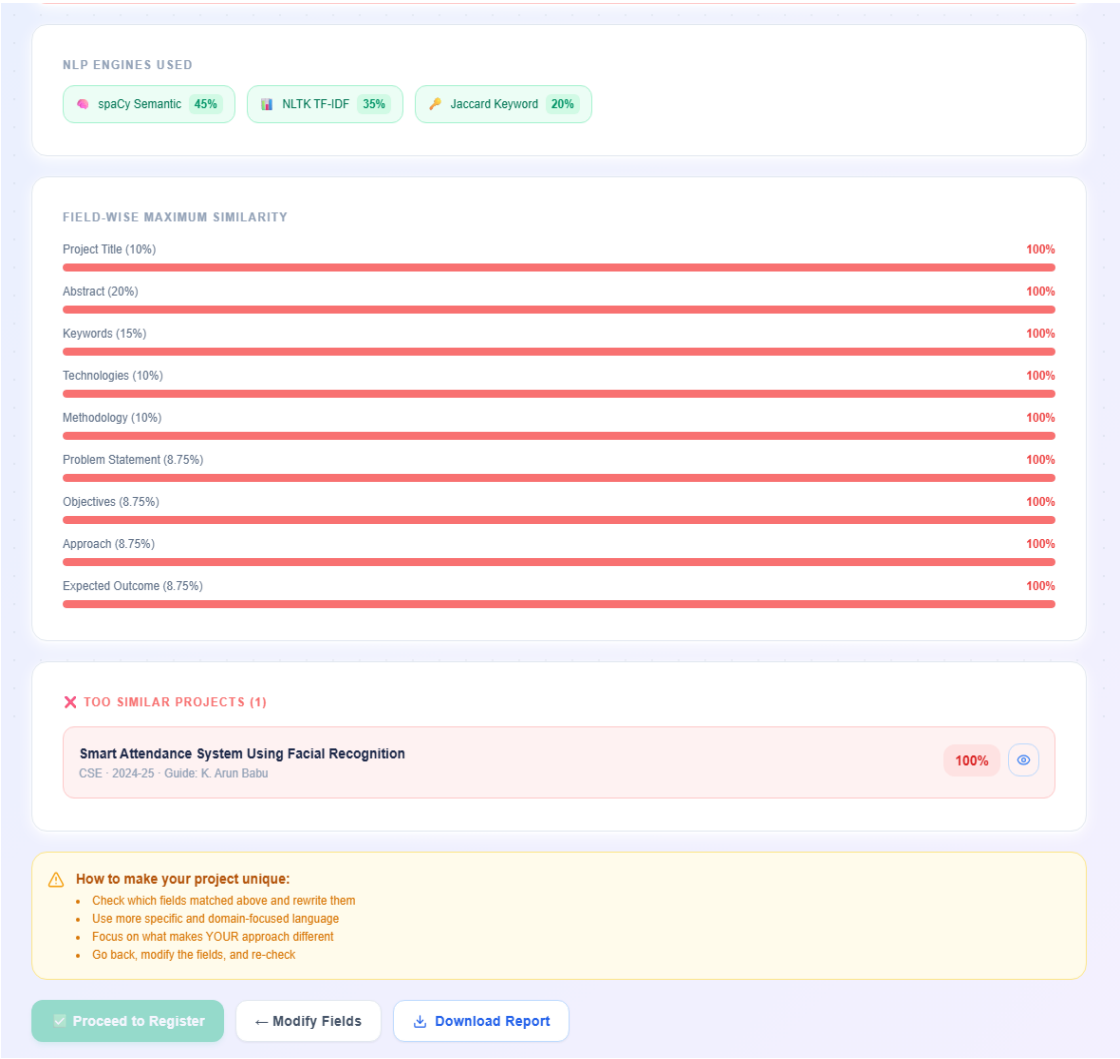
100% Objectives

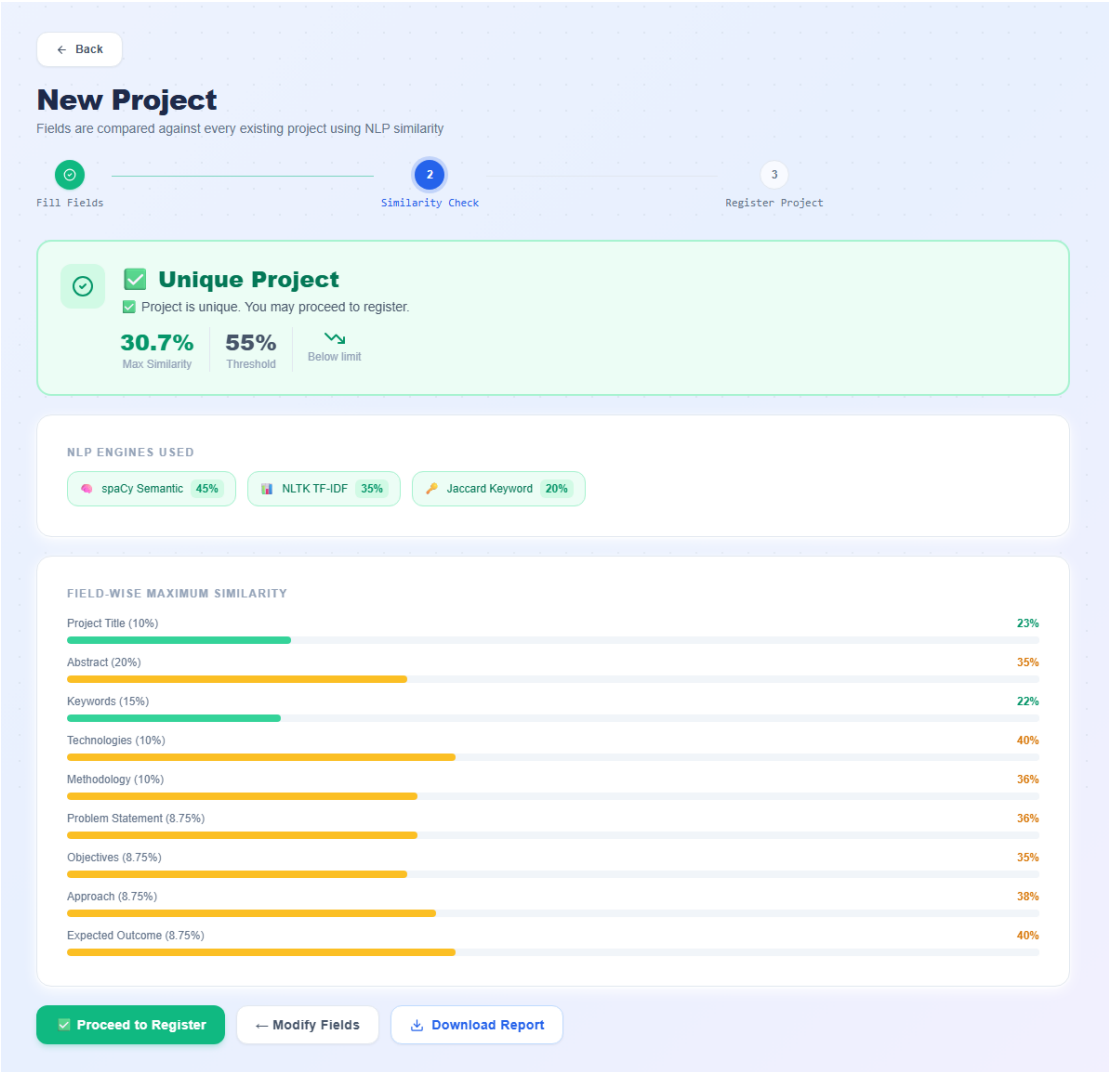
100% Approach

100% Expected Outcome

The above screen displays the Similarity Check Results of the Academic Project Selection System. It shows that the submitted project is flagged as “Too Similar”, indicating a high similarity score with an existing project. The system provides an overall similarity percentage along with a breakdown of matched fields such as title, abstract, keywords, technologies, and methodology.

Additionally, the page highlights the most similar project and compares the user's content with the matched content for better understanding. This helps users identify duplicate areas and make necessary modifications before final submission. The detailed comparison ensures transparency and supports users in improving the originality of their project.





This screen shows Step 2 of the project submission process, where the system performs an NLP-based similarity check. After the user submits project details, the system analyzes the content and displays a similarity score. In this case, the project is marked as unique since the score is below the threshold, allowing the user to proceed with registration.

The similarity is calculated using multiple techniques such as semantic analysis, TF-IDF, and keyword matching. The system also provides field-wise similarity percentages with colour indicators to show safe, acceptable, or high similarity levels.

Users are given options to proceed with registration, modify the content, or download the similarity report.

📊 ProjectVault — Similarity Report

Generated: 3/4/2026, 4:43:36 pm

Project Information

Field	Content
Title	AI-Based Academic Project Selection and Validation System
Keywords	Academic Projects, NLP, TF-IDF, Cosine Similarity, AI Validation, FastAPI, React, PostgreSQL
Technologies	Frontend: React, Tailwind CSS Backend: FastAPI, Python Database: PostgreSQL Machine Learning: Scikit-learn, NumPy NLP: spaCy, TF-IDF AI Integration: Groq API Deployment: Vercel, Render

Verdict

✅ Unique Project

30.7% Max Similarity **55%** Threshold

NLP Engines Used

Engine	Weight	Status
🧠 spaCy Semantic	45%	✅ Active
📊 NLTK TF-IDF	35%	✅ Active
🔑 Jaccard Keyword	20%	✅ Active

Field-wise Maximum Similarity

Field	Similarity	Bar
Project Title (10%)	23%	
Abstract (20%)	35%	
Keywords (15%)	22%	
Technologies (10%)	40%	
Methodology (10%)	36%	
Problem Statement (8.75%)	36%	
Objectives (8.75%)	35%	
Approach (8.75%)	38%	
Expected Outcome (8.75%)	40%	

Register Project Page

← Back

New Project

Fields are compared against every existing project using NLP similarity

Fill Fields — Similarity Check — 3 Register Project

Project passed similarity check!
Now fill in team and academic details to register.

Team & Academic Details

TEAM MEMBERS (COMMA-SEPARATED)

Alice, Bob, Charlie

GUIDE / SUPERVISOR

Prof. Dr. Smith

BRANCH / DEPARTMENT

Computer Science & Engineering

ACADEMIC YEAR

2025-26

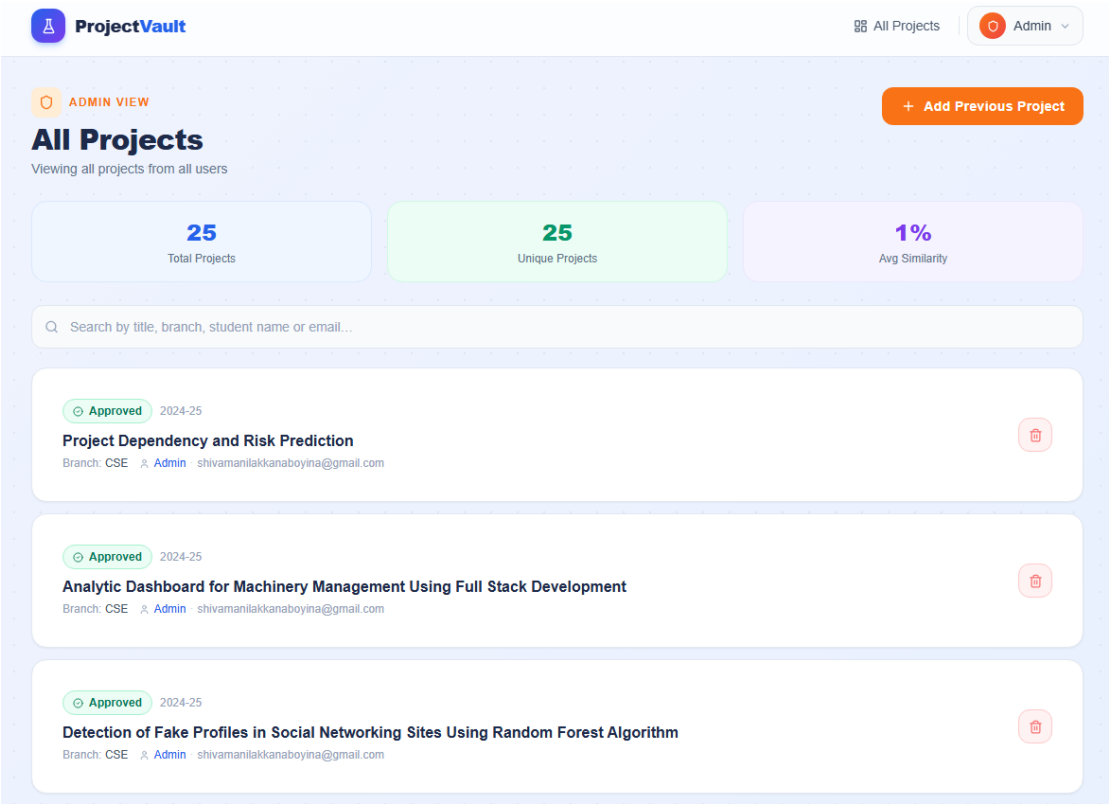
💾 Save to Database ← Back

The screen shows Step 3 of the New Project submission — the Team and Academic Details page. After passing the similarity check, the system displays a green banner stating Project passed similarity check, confirming that the project is unique and ready for registration.

At this stage, the student is required to fill in additional details such as team members (entered as comma-separated values), guide or supervisor name, branch or department, and academic year selected from a dropdown list.

Once all the required information is provided, clicking the Save to Database button stores the project permanently in the PostgreSQL database with its status set to Approved. The Back button allows the student to return to Step 2 to review the similarity results before final submission.

Admin Dashboard



The screen shows the Admin Dashboard of Project Vault, which displays all submitted projects in the system. It provides an overview of key statistics such as total projects, unique projects, and average similarity, indicating the overall originality of submissions.

The dashboard includes a search feature that allows the admin to quickly find projects based on title, branch, student name, or email. Each project entry shows details like approval status, academic year, project title, branch, and owner information.

Additionally, the admin can delete projects if needed and use the Add Previous Project option to insert older projects directly into the database.

5.1.6 Result Analysis

The Academic Project Selection System was tested and analysed to evaluate its effectiveness, accuracy, and usability. The system successfully demonstrated its ability to manage academic project submissions, detect duplicate or similar projects, and provide quality feedback through AI validation.

During testing, multiple project submissions were processed to assess the NLP similarity detection algorithm. The TF-IDF and cosine similarity methods accurately calculated similarity scores, with projects exceeding the 70% threshold being flagged as duplicates. Detailed field-wise similarity scores allowed administrators and guides to identify specific areas of overlap, ensuring originality and minimizing redundancy.

The AI-based project validation algorithm also performed effectively, evaluating project quality, completeness, and clarity. Students received actionable feedback that helped them improve their submissions, while guides could rely on the AI assessment to make informed decisions more efficiently.

The role-based dashboards provided clear insights into submission status, project similarities, and review history, allowing students, guides, and administrators to monitor the process seamlessly.

Performance analysis showed that the system handled multiple concurrent submissions without significant delays, thanks to asynchronous backend processing and efficient database queries.

Overall, the system improved the speed and accuracy of project evaluation, reduced manual effort for guides, and enhanced the transparency of project management.

The result analysis confirms that the Academic Project Selection System meets its objectives by providing a scalable, reliable, and intelligent platform for managing academic projects in educational institutions.

6 Testing and Validation

The Academic Project Selection System underwent rigorous testing and validation to ensure that all functionalities were working correctly, the system was reliable, and the AI/NLP algorithms provided accurate results. The testing process included unit testing, integration testing, and user acceptance testing.

Unit testing was performed on individual components such as API endpoints, authentication modules, and frontend forms to ensure each function produced the expected output. Integration testing verified seamless communication between the frontend, backend, and database, ensuring that data submitted through the frontend was accurately processed, stored, and retrieved by the backend.

The system's core features, including NLP-based similarity detection and AI project validation, were validated using multiple project submissions. The similarity detection algorithm was tested to ensure it correctly identified duplicate or highly similar projects, calculating similarity scores with high accuracy and storing detailed field-wise results.

The AI validation module was tested to confirm that project quality, completeness, and clarity were assessed correctly, and that meaningful feedback was provided to students to help improve their submissions.

Additionally, performance testing was conducted to evaluate the system's ability to handle multiple concurrent submissions without delays, leveraging asynchronous processing in the backend. Security validation was also carried out to ensure role-based access controls were enforced, and sensitive data such as passwords were securely stored using hashing techniques.

User acceptance testing involved students, guides, and administrators interacting with the system to confirm ease of use, clarity of dashboards, and reliability of feedback.

Overall, testing and validation confirmed that the Academic Project Selection System is accurate, secure, and efficient, providing a robust platform for managing

academic projects while minimizing manual effort and improving the quality and originality of submissions.

6.1 Features to be Tested

The Academic Project Selection System incorporates multiple features that were systematically tested to ensure proper functionality, reliability, and user satisfaction.

- **Authentication & Role-Based Access Control**
  - Tested secure login for students, guides, and administrators.
  - Verified appropriate access permissions for each role.
- **Project Submission Module**
  - Tested accurate submission of project details: title, abstract, keywords, methodology, technologies, and team info.
  - Confirmed backend successfully stores data in the database.
- **NLP Similarity Detection Feature**
  - Tested with multiple projects having varying textual similarity.
  - Verified correct calculation of TF-IDF vectors and cosine similarity scores.
  - Flagged projects exceeding similarity threshold.
  - Field-wise similarity reporting validated for transparency.
- **AI-Based Project Validation Feature**
  - Tested Groq API for analysing project quality, completeness, and clarity.
  - Ensured meaningful feedback and scores are returned to help students improve submissions.

- **Dashboard & User Interface**
    - Verified students, guides, and administrators can view project lists, status updates, similarity scores, and AI feedback accurately.
- **API Functionality & Data Integration**
    - Tested smooth communication between frontend, backend, and database.
- **Performance, Concurrency & Security**
    - Validated the system can handle multiple users simultaneously.
    - Ensured protection of sensitive information.
    - Confirmed consistent performance under load.
- **Overall Outcome**
    - System operates efficiently.
    - Provides accurate evaluations.
    - Meets functional requirements of all users.

6.2 Features not to be Tested

Testing of the Academic Project Selection System focused on verifying core functionalities, there are several features that were intentionally excluded from testing due to scope limitations, resource constraints, or dependency on external systems.

- **Advanced AI Customization**
    - Modifying Groq AI model parameters, training with additional datasets, or customizing scoring algorithms were not tested.
    - Considered future enhancements outside the current project scope.
- **Third-Party Service Updates**
    - Potential updates to hosting platforms like Vercel or Render were not tested.

- Changes in PostgreSQL database engine or external APIs like Groq were excluded.
- Testing not feasible as these are controlled externally.

- **Extreme Performance & Stress Scenarios**
  - Not tested for extremely high loads, e.g., thousands of simultaneous users or massive datasets.
  - Specialized infrastructure required; unnecessary for intended academic deployment scale.
- **Mobile & Offline Functionality**
  - Mobile-specific interface testing and offline mode were not performed.
  - System primarily designed for desktop/laptop use with active internet connectivity.
- **Future Extensibility Features**
  - Multi-language support, advanced analytics dashboards, and integration with additional AI engines were not tested.
  - Planned for later iterations of the system.
- **Focus of Testing**
  - Testing concentrated on core functionalities:
    - Project submission
    - User authentication
    - AI validation
    - NLP similarity detection
    - Dashboard performance
  - Ensured reliability, accuracy, and efficiency for primary purpose while acknowledging limitations for future improvements.

6.3 Testing Tools and Environment

The testing of the Academic Project Selection System was carried out in a controlled environment using a combination of software tools and hardware setups to ensure accuracy, reliability, and reproducibility of results.

- **Backend & API Testing**
  - Used **Postman** to validate REST API endpoints.
  - Checked response times and verified authentication mechanisms.
  - Ensured proper data handling for project submissions, similarity detection, and AI validation.
- **Unit Testing**
  - Conducted with **PyTest** for Python modules, including NLP engine and AI validator.
  - Automated testing of individual functions to ensure correct execution of algorithms.
- **Frontend Testing**
  - Used **React Testing Library** and **Jest** to verify UI components, navigation routes, and forms.
  - Ensured role-based dashboards displayed accurate information.
  - Validated form submissions before sending to the backend.
  - Used browser developer tools to monitor network requests, console logs, and page performance.
- **Hardware Environment**
  - Standard computing devices with Intel i3 or higher processors.
  - 4–8 GB RAM and SSD storage to ensure smooth execution of frontend and backend operations.
- **Software Environment**
  - Python 3.11, FastAPI 0.115.0 for backend.
  - React 18+ for frontend.
  - PostgreSQL 16 for database management.
- **Cloud Hosting**
  - **Vercel** for frontend hosting.
  - **Render** for backend and database hosting.

- Environment variables (e.g., `GROQ_API_KEY`, `DATABASE_URL`) used for secure API access.

- **Testing Objectives & Outcomes**
  - Enabled functional, integration, performance, and security testing.
  - Ensured reliable operation under normal academic workloads.
  - Provided accurate, secure, and user-friendly results.

6.4 Design of test cases and scenarios

Test Case ID	Test Scenario	Input Data	Expected Output	Result
TC01	User Registration	Valid email & password	User account created successfully	Pass
TC02	User Login	Correct credentials	Login successful	Pass
TC03	Invalid Login	Wrong password	Error message displayed	Pass
TC04	Project Submission	Complete project details	Project saved in database	Pass
TC05	Project Submission Validation	Missing mandatory fields	Validation error shown	Pass
TC06	NLP Similarity Detection	Submit projects with duplicate/partial/unique content	Correct similarity scores; duplicates flagged appropriately	Pass

Test Case ID	Test Scenario	Input Data	Expected Output	Result
TC07	AI-Based Project Validation	Submit projects with complete/incomplete content	Feedback and quality scores returned	Pass
TC08	Dashboard Access	Logged in student/guide/admin	Role-based dashboard displayed with relevant data	Pass
TC09	Role-Based Access Control	Attempt unauthorized access	Access denied for unauthorized actions	Pass
TC10	Performance & Concurrency	Multiple simultaneous submissions	System handles concurrent users efficiently	Pass

7 Conclusion and Future Enhancements

In conclusion, the Academic Project Selection System successfully achieves its primary objective of providing a reliable, efficient, and intelligent platform for managing academic project submissions. By integrating NLP-based similarity detection and AI-powered project validation, the system ensures that submitted projects are original, high-quality, and meet academic standards. Role-based dashboards and seamless frontend-backend interaction enhance usability for students, guides, and administrators, while secure authentication and data storage ensure privacy and integrity. The system demonstrates efficient handling of multiple submissions, accurate similarity scoring, and meaningful AI feedback, significantly reducing manual effort and improving transparency in project evaluation.

For future enhancements, several improvements can be considered to further increase the system's effectiveness and scalability. These include mobile application support to allow students and guides to access the system on smartphones and tablets, multi-language support for broader accessibility, and advanced analytics dashboards for administrators to visualize submission trends, similarity patterns, and performance metrics. Integration with additional AI models could provide more detailed quality assessments, including plagiarism detection across external sources and advanced methodology evaluation. Furthermore, implementing bulk project import/export features, notifications and alerts, and offline submission support would enhance convenience and usability. Overall, the Academic Project Selection System provides a solid foundation for modern academic project management, with ample opportunities for future growth and refinement to meet evolving educational needs.

In addition to the above, the system can be further enhanced by incorporating real-time collaboration features, allowing students and guides to communicate directly within the platform through chat or discussion threads. This would improve interaction and reduce dependency on external communication tools. The integration of automated email or SMS notifications for important events such as project approval, rejection, or feedback updates would ensure timely communication and improve user engagement.

Another potential improvement is the implementation of advanced security mechanisms, such as multi-factor authentication and role-based encryption, to further

strengthen data protection. The system can also be extended to support cloud-based scalability, enabling it to handle larger datasets and a higher number of users across multiple institutions.

Additionally, incorporating version control for project submissions would allow students to track changes and maintain a history of updates, improving transparency and accountability. Finally, the inclusion of data analytics and reporting tools can help institutions gain valuable insights into project trends, student performance, and research areas.

Features like graphical reports, downloadable summaries, and predictive analysis can assist administrators in decision-making and academic planning. With these enhancements, the Academic Project Selection System can evolve into a more comprehensive, scalable, and intelligent platform, capable of supporting the growing demands of modern educational environments.

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